STAT 306 Group Project (Group P14) — Draft 1

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Introduction

This report investigates the relationship between economic development, industrialization, and cereal crop yields across countries from 2019 to 2024 using multiple linear regression. Specifically, we ask: *to what degree do GDP per capita and CO_2 emissions per capita explain cereal crop yields when controlling for continent?*

GDP per capita serves as a proxy for economic development and CO_2 emissions per capita as a proxy for industrialization. Continent is included as a categorical control covariate to account for broad regional differences not captured by the two continuous predictors.

All data were obtained from Our World in Data, which aggregates data from the Food and Agriculture Organization (FAO) and the World Bank. Three datasets were used: cereal crop yields (tonnes per hectare), GDP per capita in constant international dollars, and CO_2 emissions per capita in tonnes per person. Data were available as annual time series from approximately 1990 to 2024. To avoid time series modelling while retaining meaningful representation, we aggregated each variable over the 2019–2024 period by computing country-level means. This five-year window also helps normalize short-term confounders such as weather variability.

Analysis

Data Preparation

The three datasets were filtered to the years 2019–2024 and averaged per country, producing a single cross-sectional observation per country. The three datasets were joined by country name. Continent Regional aggregates (e.g., "World", "High-income countries") that were not matched to a specific country were removed. Countries with missing values were also removed. This left 170 countries, with one notable outlier, Cape Verde (Figure 3). Cape Verde was removed from the model due to extreme structural challenges that drastically reduced its cereal yield. These factors include a low dependence on the agriculture industry and an unstable climate (Brilhante et al., 2021). As these factors are outside of the scope of this model, this country was removed. Therefore, the final data set contained 169 countries on five continents: Africa (50), Asia (44), Europe (38), the Americas (31), and Oceania (6).

Exploratory Data Analysis

Summary statistics revealed that all three numeric variables (cereal yield, GDP per capita, and CO_2 per capita) were right-skewed in their raw form, with means notably exceeding their medians

(Table 1). Log-transforming all three variables produced approximately symmetric, bell-shaped distributions, motivating a fully log-log model specification. Additionally, models fitted using log transformed variables yielded notably better fits, in terms of residual spread and Adjusted R^2 (Figure 2).

Table 1: Summary statistics for the three numeric variables (n = 169).

Variable	Min	Median	Mean	Max
Cereal yield (t/ha)	0.02	3.26	3.78	21.72
GDP per capita (Int'l $)	1,040	15,400	23,492	132,286
CO_2 per capita (t/person)	0.05	2.45	4.36	37.44

Scatterplots of log yield against log GDP per capita and log CO_2 per capita revealed clear positive linear relationships, confirming that the log-log scale was appropriate for modelling. Colour-coding by continent revealed strong clustering: European countries tended to occupy the high-GDP, high-CO_2, high-yield region, while African countries clustered in the opposite corner. This visual pattern indicated that continent was a meaningful control variable that needed to be included in the model.

A boxplot of log yield by continent (Figure 1) confirmed substantial between-continent differences. Africa had the lowest median log yield and the widest spread, while Europe had the highest and most consistent yields. The Americas, Asia, and Oceania fell in a similar middle range. Oceania showed wide spread, likely due to having only six countries.

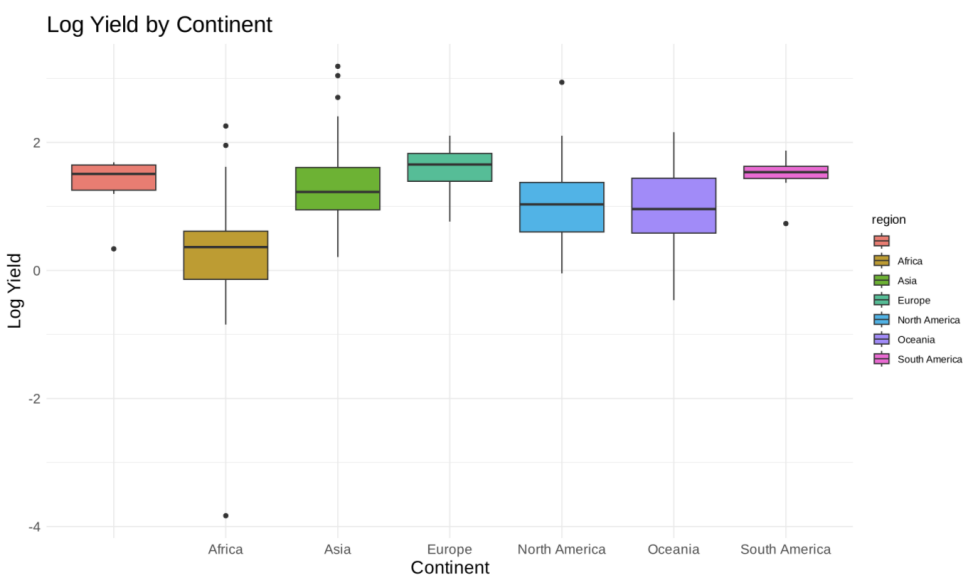

Figure 1: Log cereal yield by continent. Africa is the reference group. Oceania has only 6 countries, leading to wide uncertainty.

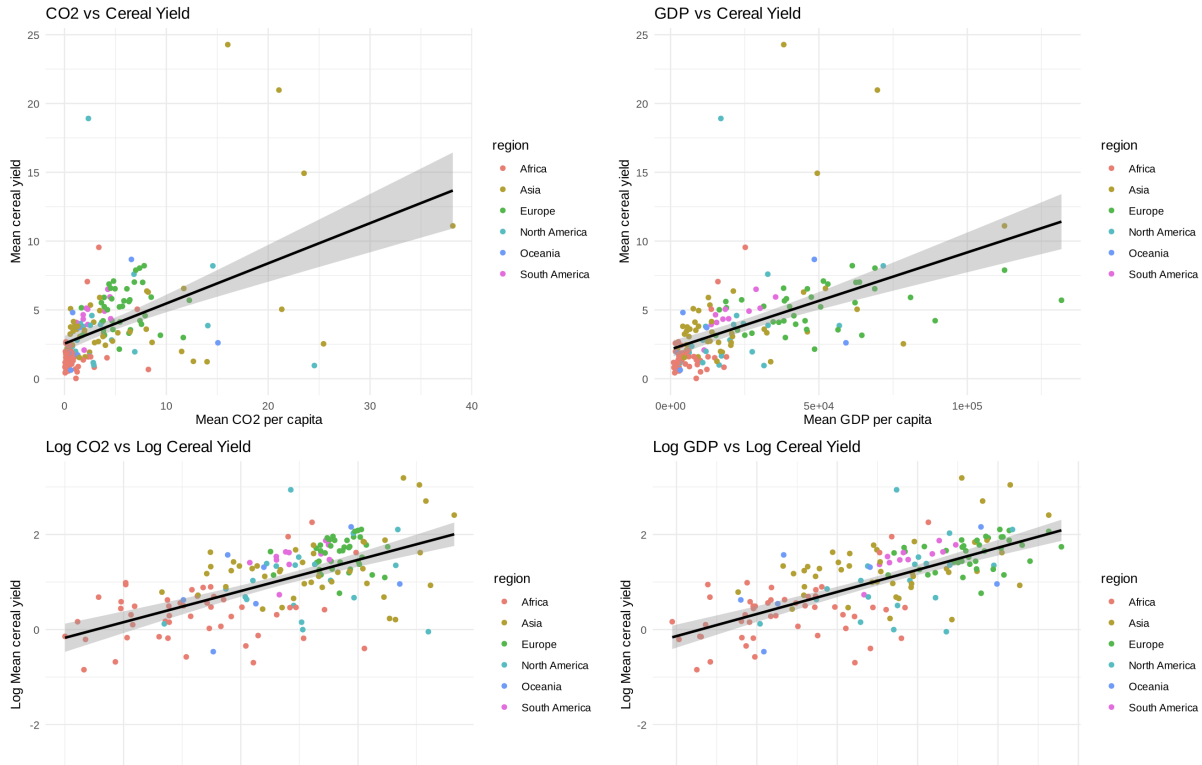

Figure 2: Cereal yield against CO2 per capita (top left) and GDP per capita (top right), and log cereal yield against log CO2 per capita (bottom left) and log GDP per capita (bottom right), coloured by continent. Both predictors show a positive linear trend with yield on the log-log scale.

Multicollinearity. The Pearson correlation between log GDP per capita and log CO_2 per capita was $r = 0.911$, indicating very high collinearity between the two main predictors. This was expected, as wealthier countries tend to emit more CO_2. Given this near-linear relationship between predictors, we fit an additive model rather than an interaction model between GDP and CO_2, as an interaction term would further inflate multicollinearity and make coefficients uninterpretable.

Outlier. One extreme observation was identified: Cape Verde, a small island nation off the coast of West Africa, had a mean cereal yield of 0.019 tonnes per hectare (log yield ≈ -4), far below any other country. This reflects Cape Verde's near-total lack of arable land rather than a data error.

Model

We fit a multiple linear regression model with log cereal yield as the response and log GDP per capita, log CO_2 per capita, and continent (with Africa as the reference level) as predictors:

$$\begin{aligned}\log(\text{yield}_i) = \beta_0 &+ \beta_1 \log(\text{GDP}_i) + \beta_2 \log(\text{CO}_{2i}) \\ &+ \beta_3 \text{Asia}_i + \beta_4 \text{Europe}_i + \beta_5 \text{NorthAmerica}_i \\ &+ \beta_6 \text{Oceania}_i + \beta_7 \text{SouthAmerica}_i + \varepsilon_i\end{aligned}$$

An initial model fit on all 170 countries showed Cape Verde to be a highly influential observation,

lying well beyond the Cook's distance thresholds in the Residuals vs. Leverage diagnostic plot. Given that Cape Verde is not representative of typical agricultural countries, it was removed and the model was re-fit on the remaining 169 countries.

Model Diagnostics

Diagnostic plots for the final model (Figure 3) showed no strong non-linear pattern in the Residuals vs. Fitted plot, suggesting the log-log specification was appropriate. The Q-Q plot showed reasonable normality across most of the distribution, with mild deviation in the upper and lower tail.

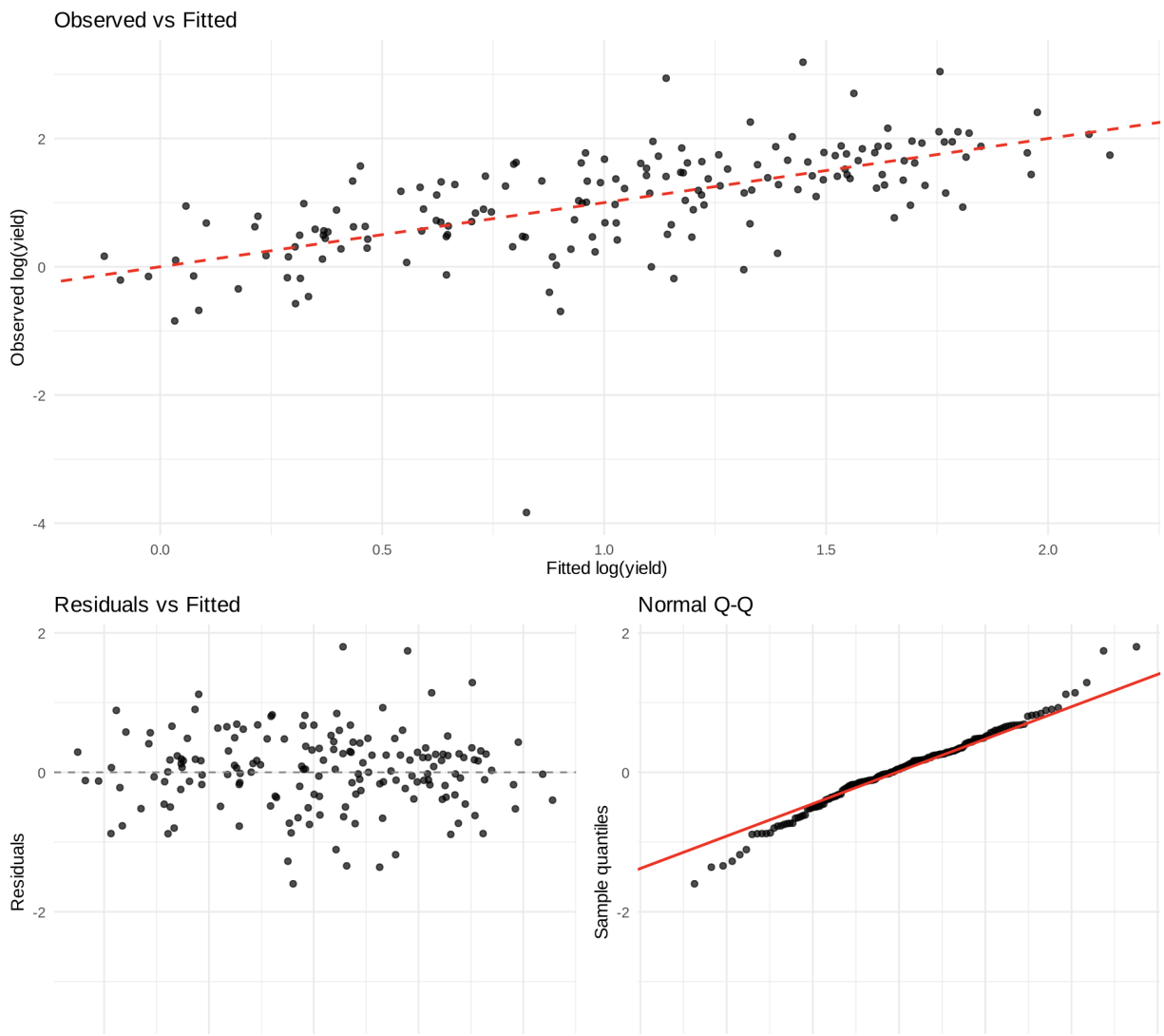

Figure 3: Regression diagnostic plots for the final model (Cape Verde removed, n=169). Clockwise from top-left: Observed vs. Fitted, Residuals vs. Fitted, Normal Q-Q.

Results

The final model results are presented in Table 2. The model explained 54.9% of the variation in log cereal yield (adjusted $R^2 = 0.530$, $F(7, 161) = 28.03$, $p < 0.001$).

Table 2: Multiple linear regression results. Response: log(mean cereal yield). Reference region: Africa. $n = 169$.

Term	Estimate	Std. Error	t-value	p-value
Intercept	-4.315	0.944	-4.57	< 0.001
log(GDP per capita)	0.540	0.106	5.10	< 0.001
log(CO_2 per capita)	-0.178	0.078	-2.28	0.024
Asia (vs. Africa)	0.631	0.133	4.74	< 0.001
Europe (vs. Africa)	0.443	0.165	2.69	0.008
North America (vs. Africa)	0.266	0.157	1.70	0.091
Oceania (vs. Africa)	0.450	0.216	2.09	0.039
South America (vs. Africa)	0.633	0.191	3.32	0.001
$R^2 = 0.549$, Adjusted $R^2 = 0.530$, RSE $= 0.519$, $p < 0.001$				

Conclusion

This analysis investigated the degree to which GDP per capita and CO_2 emissions per capita explain cereal crop yields across 169 countries from 2019 to 2024, controlling for continent.

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GDP per capita was the strongest and most reliable predictor. A 1% increase in GDP per capita was associated with a 0.540% increase in cereal yield, holding CO_2 emissions per capita and continent constant. This is consistent with the idea that wealthier countries have greater access to agricultural technology, mechanization, fertilizers, and irrigation, all of which drive higher land productivity.

CO_2 per capita showed a statistically significant negative association with yield ($\hat{\beta} = -0.178$, $p = 0.024$). This counterintuitive sign should be interpreted cautiously. It is most likely a consequence of the severe multicollinearity between GDP and CO_2 ($r = 0.911$, VIF $= 7.718$). Once GDP is in the model, CO_2 no longer independently captures industrialization or development; instead, it may reflect residual variation from countries that emit heavily relative to their economic output with less efficient agricultural sectors. This coefficient should not be interpreted causally.

Continent was a meaningful control variable. All continents yielded significantly more than Africa (the reference) except North America, which was borderline ($p = 0.091$). This confirms that continent-level factors (soil quality, climate, historical agricultural investment, and institutional capacity) explain substantial variation in yield beyond what GDP and CO_2 alone capture.

Limitations. The most significant limitation is the high multicollinearity between log GDP and log CO_2 (VIF > 5 for both), which prevents the model from cleanly isolating their individual effects. Oceania is underrepresented with only six countries, limiting the precision of that coefficient.

The cross-sectional design means causality cannot be inferred, as associations found here are correlational. Finally, important drivers of yield such as fertilizer use, irrigation access, arable land quality, climate, and agricultural policy are omitted from the model and likely correlate with GDP, meaning the GDP coefficient may partially capture their effects.

Overall, the results provide evidence that economic development is a key correlate of cereal crop yield efficiency, though the high collinearity between predictors limits the degree to which their individual contributions can be disentangled. Future work could address multicollinearity by using principal component regression or by substituting one of the correlated predictors with a more orthogonal measure of industrialization.

Code Submission

Evan Barr submitted the R code and data sets used in this project.

Statement of AI Use

Generative AI (Claude, Anthropic) was used in this project to assist with the following tasks: setting up the R coding environment and GitHub repository, writing and debugging R code for data cleaning, exploratory data analysis, and regression modelling. All statistical decisions (including the choice of log transformation, the selection of Africa as the reference continent, the removal of Cape Verde, and the choice of an additive over an interactive model) were made by us based on the outputs and guided by course material.

Appendix

AI Prompts Used

- Prompt 1: Debug the following R code for merging datasets and computing country-level averages.
- Prompt 2: How do I compute Cook's distance and standardized residuals in R for a linear regression model?
- Prompt 3: How can I identify influential observations and determine whether an observation should be removed from a regression model?
- Prompt 4: Help format regression output into a LaTeX table with estimates, standard errors, and p-values.
- Prompt 5: Fix LaTeX formatting issues for equations and tables so that they do not go off the page.
- Prompt 6: Debug ggplot and base R code

References

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